

ABDULLAH BASIT

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PROFESSIONAL SUMMARY

Electrical Engineering (Computer Engineering) graduate from COMSATS University Islamabad with strong full-stack development experience and a practical AI/ML portfolio. Designed and delivered 7 production-ready software projects spanning web applications, desktop systems, AI automation platforms, and deep learning pipelines. Proficient in the MERN stack, REST API design, JWT authentication, SQL/NoSQL databases, and cloud deployment (Render, Vercel, MongoDB Atlas). Experienced integrating large language models (Anthropic Claude, OpenAI GPT-4o) into multi-channel automation systems. Seeking a Software Engineering, Full-Stack, or AI Engineering role to contribute scalable, maintainable software solutions.

EDUCATION

BS Electrical Engineering (Computer Engineering)

Jan 2026 — Graduated

COMSATS University Islamabad

Relevant Coursework: Data Structures & Algorithms · Computer Networks · Cloud Computing · Artificial Intelligence · Digital Logic Design · Database Systems

EXPERIENCE

RF & Drive Test Intern — TalkPool LCC (Huawei Projects)

April 2026 – June 2026

- Conducted RF drive tests across Islamabad using TEMS Investigation to collect LTE signal quality data (RSRP, RSRQ, SINR) for network performance evaluation and optimization
- Processed and analyzed drive test data using Genex Assistant; identified coverage gaps and submitted RF optimization findings to the project team
- Studied GSM, WCDMA, and LTE network architecture, RF planning concepts, and network sharing technologies (MORAN, MOCN)

PROJECTS

AI Omnichannel Auto-Reply Automation System | github.com/abdu2256

Node.js · Express.js · MongoDB · React.js · Anthropic Claude API · OpenAI GPT-4o · Twilio · Nodemailer · JWT · CI/CD

- Architected and deployed a production-grade SaaS automation platform that routes AI-generated replies across Email, WhatsApp, and Slack channels using a confidence-based dual-provider model (Claude + GPT-4o)
- Engineered real-time streaming responses via Server-Sent Events (SSE) and implemented intent detection, sentiment analysis, multi-language support (English/Urdu/Roman Urdu), and human handoff detection
- Designed 15+ RESTful API endpoints with MongoDB schemas (Message, Channel, ActivityLog), role-based access control, JWT authentication, and bcrypt password hashing
- Integrated Twilio WhatsApp Business API, Gmail SMTP/IMAP with node-cron scheduling, and Slack Events API for multi-channel orchestration
- Deployed backend on Render and React frontend on Vercel with GitHub CI/CD pipeline; configured MongoDB Atlas M0 for cloud persistence

AI Assistant Chatbot

React.js · Node.js · Express.js · Anthropic Claude API · OpenAI GPT-4o · MongoDB · JWT · WebSocket · SSE

- Built a full-stack conversational AI assistant with real-time streaming responses via WebSocket/SSE and dynamic model switching between Claude and GPT-4o per user preference
- Implemented persistent conversation history, JWT-authenticated sessions, multi-language support, and an admin analytics dashboard tracking usage and response quality metrics
- Deployed on Render (backend) and Vercel (frontend) with CI/CD via GitHub and MongoDB Atlas for cloud data storage

Clinic Management System

Electron.js · Node.js · Express.js · React.js · SQLite · JWT · REST API

- Developed a cross-platform offline-first desktop application using Electron.js for clinic operations on Windows, macOS, and Linux
- Built appointment scheduling, patient registration, and medical history modules with a relational SQLite database schema covering patients, appointments, doctors, and billing records
- Implemented a billing and invoicing module with payment status tracking, fee calculation logic, and printable receipt generation
- Packaged the application as a standalone desktop installer embedding a local Express.js REST API backend rendered within Electron

TalkPool KPI Analytics Dashboard

React.js · Recharts · Node.js · Express.js · MongoDB · REST API · JWT

- Designed an interactive telecom KPI dashboard visualizing LTE network performance metrics (RSRP, RSRQ, SINR, throughput) using Recharts line, bar, and heatmap components
- Built configurable reporting module with date-range filtering, threshold-based alerting, and exportable KPI reports for field engineers and NOC teams
- Implemented MongoDB aggregation pipelines for efficient KPI data querying; secured routes with JWT role-based authentication

University Management Portal — CUOnline Style

Node.js · Express.js · MongoDB · React.js · JWT · bcrypt · REST API · Mongoose

- Delivered a full-stack university portal with role-based dashboards (Student, Faculty, Admin) modeled on COMSATS CUOnline workflows
- Built 7 Mongoose models (User, Course, Enrollment, Attendance, Marks, Fee, FYP) and 25+ JWT-protected REST endpoints; seeded complete demo database for full-feature testing
- Student module: real-time attendance tracking, result cards, fee statements, course registration, FYP tracking, timetable; Faculty module: mark attendance, upload grades for assignments/quizzes/exams

Skin Disease Detection System Using CNN (Final Year Project)

Python · TensorFlow · Keras · OpenCV · Deep Learning · Image Processing

- Designed and trained a CNN-based multi-class classifier for skin disease detection from medical images, achieving strong classification accuracy on held-out test data
- Applied image preprocessing pipeline (normalization, augmentation, resizing) using OpenCV and TensorFlow/Keras to address class imbalance and improve model generalization

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), HTML5, CSS3, SQL, Bash (Linux)

Frontend: React.js, React Router, Recharts, Lucide React, Electron.js, Responsive UI Design

Backend: Node.js, Express.js, REST API Design, JWT, bcrypt, Nodemailer, WebSocket, SSE

Databases: MongoDB, Mongoose ODM, SQLite, NoSQL Data Modeling, SQL Queries

AI / ML: TensorFlow, Keras, CNN, OpenCV, Anthropic Claude API, OpenAI GPT-4o API, Prompt Engineering, LLM Integration

Cloud & DevOps: Git, GitHub, Vercel, Render, MongoDB Atlas, CI/CD (GitHub Actions basics), Linux CLI, ngrok, System Monitoring

Messaging / APIs: Twilio WhatsApp API, Gmail SMTP/IMAP, Slack Events API, RESTful Integration

Telecom Tools: TEMS Investigation, Genex Assistant, LTE/4G RF KPI Analysis, Drive Testing

Other: MERN Stack, MVC Architecture, Agile basics, Application Troubleshooting, Desktop App Development

CERTIFICATIONS & TRAINING

- RF Telecom Training — Huawei Technologies, Islamabad (May 2026 – Present)
- RF & Drive Test — TalkPool LCC (Huawei Projects), Islamabad (April 2026 – June 2026)
- MERN Stack Development — Self-directed, delivered through production project portfolio
- AI & Deep Learning Fundamentals — Applied through Final Year Project and independent AI projects